

Module 2 [Constructor, const & static] (for Week 2, Based on CO1)

1. **Problem Statement:** Use the following class Test with the data members and methods

```
class Test{
private:
    Data Member x;
    Data Member y;
    Data Member z;
public:
    //write methods

};
```

Code:

```
#include<iostream>
using namespace std;
class Test
{
private:
    int x;
    int y;
    static int z;

public:
    Test()
    {
        x=0;
        y=0;
        z++;
    }

    Test(int a, int b)
    {
        x=a;
        y=b;
        z++;
    }

    Test(const Test &obj)
    {
        x = obj.x;
        y = obj.y;
        z++;
    }

    void setxy(int a, int b)
    {
```

```

        x=a;
        y=b;
    }
    void display() const
    {
        cout << x << y << z << endl;
    }
    int getx()
    {
        return x;
    }
    int gety()
    {
        return y;
    }
    ~Test()
    {
        cout << "Destroyed" << endl;
    }
};

int Test::z=0;
int main()
{
    int y[10];
    Test t1;
    Test t2(5,4);
    Test t3(t2);
    Test t4,t5;

    t4.setxy(10,20);
    t5.setxy(25,14);
    int sumx = t1.getx()+t2.getx()+t3.getx()+t4.getx()+t5.getx();
    y[0]=t1.gety();
    y[1]=t2.gety();
    y[2]=t3.gety();
    y[3]=t4.gety();
    y[4]=t5.gety();
    int max = y[0];
    for(int i=0; i<5; i++)
    {
        max=(max>y[i]?max:y[i]);
    }
    cout << "Sum of X: " << sumx << endl;
    cout << "Largest Y: " << max << endl;
    return 0;
}

```

```

PS D:\Satu\1-2\CSE 1204\week2> cd "d:\Sa
+ lab2.cpp -o lab2 } ; if ($?) { .\lab2
Sum of X: 45
Largest Y: 20
Destroyed
Destroyed
Destroyed
Destroyed
Destroyed
PS D:\Satu\1-2\CSE 1204\week2>

```

Now write and necessary codes in class and **main()** do the following

- i) Initialize private data members **x** and **y** to 0 when empty constructor is called
- ii) Initialize private data members **x** and **y** using parameterized constructor is called
- iii) Initialize private data members **x** and **y** from another object using copy constructor
- iv) The data member **z** keeps track of total objects created
- v) Write a method to initialize **x** and **y**
- vi) Write a method to display data member **z** only
- vii) Write a method to display **x, y** and **z** where their values can't be changed
- viii) Create five objects
- ix) Find the sum of **x**
- x) Find the object number whose **y** value is maximum
- xi) Write destructor

2. Problem Statement : Write a class A with two data member **a** and **b**. Now set the values of **a** and **b** of an object **obj1** by a setter method **SetData()** and copy the values of **a** and **b** of **obj1** to another object **obj2** by using copy constructor.

```

#include<iostream>
using namespace std;
class A
{
    int a;
    int b;

```

```

public:
    A()
    {

    }
    void setdata(int x, int y)
    {
        a=x;
        b=y;
    }
    A(const A &obj)
    {
        a=obj.a;
        b=obj.b;
    }
    void display()
    {
        cout << a << " " << b << endl;
    }
};

int main()
{
    A obj1;
    obj1.setdata(13,34);
    A obj2(obj1);
    cout << "Object 1: " ; obj1.display();
    cout << "Object 2: " ; obj2.display();

    return 0;
}

```

Output:

```

PS D:\Satu\1-2\CSE 1204\week2> cd
+ prblm2.cpp -o prblm2 } ; if ($?
Object 1: 13 34
Object 2: 13 34
PS D:\Satu\1-2\CSE 1204\week2> 

```